

Movie Recommendation System

1. Objective

The objective of this project is to recommend similar movies based on user input using content-based filtering. The system analyzes movie features like genres, keywords, cast, and overview to provide accurate recommendations.

2. Tools Used

Python

Pandas

NumPy

Scikit-learn

Google Colab

Streamlit (for UI)

3. Methodology

- Data Preprocessing:

The dataset was cleaned by merging movie and credits data, removing null values, and selecting important columns.

- Feature Extraction:

Important features such as genres, keywords, cast, crew, and overview were extracted and converted into usable formats.

- Feature Engineering:

Text data was processed by removing spaces and combining all features into a single column called "tags".

- Vectorization:

Count Vectorizer was used to convert text data into numerical vectors.

- Cosine Similarity:

Cosine similarity was used to calculate similarity between movies based on feature vectors.

4. Result

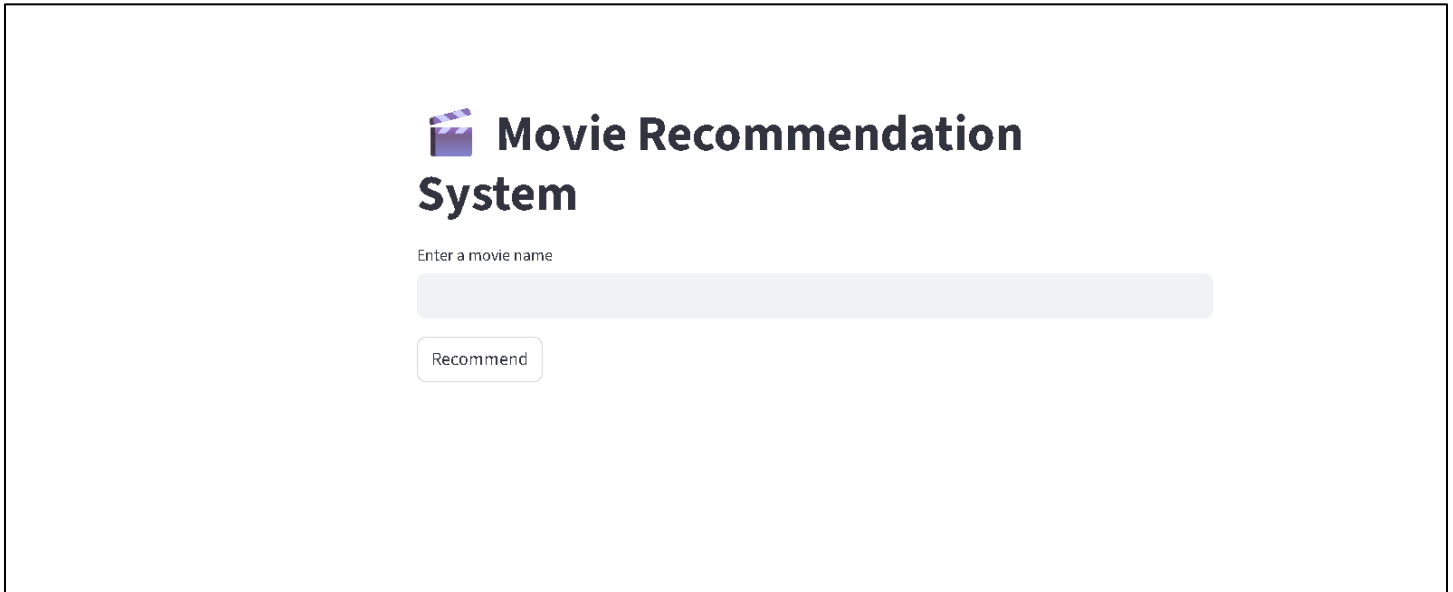
The system successfully recommends 5 similar movies for a given movie input. If the movie is not found, it shows an error message.

5. Conclusion

The movie recommendation system works efficiently using content-based filtering. It provides personalized movie suggestions and can be further enhanced using advanced machine learning techniques and a better user interface.

6. Screenshots

- **Application Interface (Initial Screen)**



 **Movie Recommendation System**

Enter a movie name

- **User Input (Entering Movie Name)**



Movie Recommendation System

Enter a movie name

Press Enter to apply

Recommend

- **Recommended Movies Output**

Movie Recommendation System

Enter a movie name

Avatar

Recommend

Titan A.E.

Small Soldiers

Ender's Game

Aliens vs Predator: Requiem

Independence Day